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Predict The Warehouse Capacity

To predict warehouse capacity, we must first verify that the mathematical model built using 80% of the data (114 data points) for training data is accurate by testing it on the remaining 20% (30 data points) for testing data. The mathematical model we obtained from the answer to question number 1 are:

Polynomial of degree 2:

$$y = 0{,}5130x^2 + 18{,}4852x + 2014{,}8884$$

Polynomial of degree 3:

$$y = 0{,}0016x^3 + 0{,}2298x^2 + 31{,}5678x + 1886{,}7842$$

Note:
y = predicted production quantity
x = time/monthly sequence

Now we will validate the model using the testing data and compare the evaluation metrics between the polynomial of degree 2 and the polynomial of degree 3.

a. Mean Absolute Error (MAE)
   Degree 2: 1.070,0909
   Degree 3: 681,3065
   The difference is very significant. On new data, the predictions from the Degree 2 model are off by about 1070 points on average, whereas the Degree 3 model's predictions are only off by about 681 points. This is no longer a small margin; the Degree 3 model is substantially almost 400 points more accurate in its average prediction.

b. Mean Squared Error (MSE)
   Degree 2: 1.500.044,91
   Degree 3: 630.863,36
   The MSE value for the Degree 2 Model is more than double that of the Degree 3 Model. This is a very large difference and indicates that the Degree 2 model likely makes some very critical prediction errors on the test data, for which it is heavily penalized by the MSE calculation.

c. Root Mean Squared Error (RMSE)
   Degree 2: 1.224,7632
   Degree 3: 794,2691
   The typical error of the Degree 2 model is very large (over 1200 points), while the Degree 3 model manages to keep its typical error under 800 points. This confirms that the Degree 2 model is prone to making large errors, whereas the Degree 3 model is much more consistent and stable.

d. R-squared (R2)
   Degree 2: 0,5686
   Degree 3: 0,8186
   The Degree 2 model's R2 of 0,5686 means it can only explain about 57% of the test data's pattern. This is a mediocre performance, leaving 43% of the pattern unexplained. In contrast, the Degree 3 model's R2 of 0,8186 means it can successfully explain 82% of the test data's pattern. This is a very strong result, showing that the model truly understands the underlying trend, even in data it has never seen before.

In conclusion, the polynomial of degree 3 performs better than the polynomial of degree 2.

The visual comparison also shows that the polynomial of degree 3 fits the data better than the polynomial of degree 2.



